SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

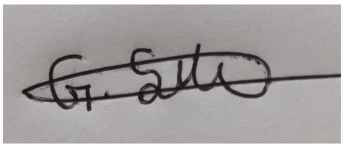
Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

*I **G.Mohammad shareef/(192311288)**_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with	Incorrect or irrelevant answer

				noticeable errors	
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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① Responsive web Design (RWD) and its importance:

Responsive web Design (RWD) is a web Development approach that allows a website to automatically adjust its layout, images, and content according to the screen size and device being used (mobile, tablet or desktop).

Importance:

* Provides a consistent user experience across all devices.
* improves website accessibility and usability
* Increases customer satisfaction and engagement.
* Enhances SEO since search engines favour mobiles.

1. Fluid Layouts:

* use relative units like %, vw, vh, em and rem instead fixed pixels.

2. Flexible images: CSS

```
img {
max-width : 100%;
height : auto;
}
```

3. Media Queries: CSS

```
@media (max-width : 786 px) {
body {
```

```
font size :16 PX;
}
}
```

Using CSS Flexbox for Adaptive Layout: HTML

Flexbox automatically arranges items according to available space.

```
<div class="container">
<div class="box">Menu</div>
<div class="box">content</div>
<div class="box">sidebar</div>
```

CSS:-

```
container
  display: flex;
  gap: 10PX;
}
box
  flex: 1;
  Padding: 20PX;
  background: light blue;
}
@media (max-width : 768PX){
  container{
flex-direction: column;
}
}
```

Improvements for Better user Experience:-

* use responsive navigation menus.
* optimize image for faster loading.
* Make buttons larger font sizes.

(22) Client-side validation and its purpose:

Client-side validation checks user input in the browser before sending data to the server.

Purpose:-

* Prevents invalid data submission.
* Provides immediate feedback.
* Reduces unecessary server requests.
* improves user experience.

Java script Event Handling in Validation:

Java script validation input using events such as:

*submit *input *change *blur

Role of Regular Expressions:-

Regular expressions (Regex) check whether input follows a required Pattern.

Email validation:-

```
let Pattern = /^[^\s@]+@^\s@]+\.[^\s@]+$/;
if Pattern.test(email)){
  alert("valid Email");
}
else{
  alert("Invalid Email");
}
```

Phone Number Validation: Java script:-

```
let phonePattern = /^[0-9]{10}$/;
```

Example Form:- HTML

```
</form id="myForm">
<input type="email" id="email">
<button>submit</button>
</form>
```

Improvements:-

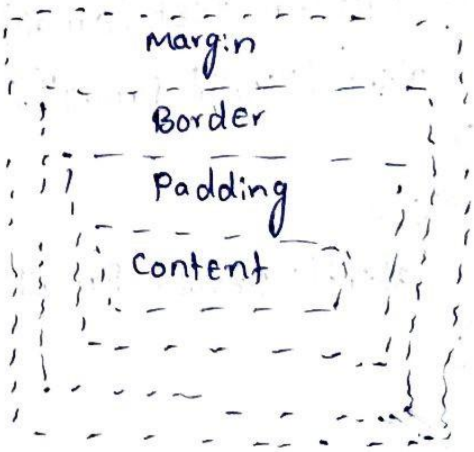
* Display error messages beside input fields.
* use HTML5 validation (required, Pattern, type = "email").
* Always perform server-side validation for security.
* sanitize user input to prevent attacks.

Q3 CSS Box Model and Layout issues:-

The CSS Box Model describes how elements are displayed on a webpage.

Components of model Box:-

1. content - Actual text or image
2. Padding - space around the conent.
3. Border - surrounds the Padding.
4. Margin - space outside the border.



CSS positioning Techniques:-

Static - Default position.

Relative- Moves relative to its normal position.

Absolute- Positioned relative to the nearest positioned ancestor.

Fixed - Fixed to the browser window.

sticky. Acts like relative until scrolling.

causes of Layout Breaking:-

*Fixed widths using pixels.

*Large image without scaling.

*incorrect positioning.

*overflowing content.

*Missing responsive design.

using Media Queries:- CSS:

```
@media (max-width: 768px){
.container{
flex-direction :column;
}
}
```

Best practices:-

*use Flexbox or Grid *Apply box-sizing: border-box.

* use Percentage widths instead of fixed pixels.

CSS:-

```
*{
boxi-sizing: border-box;
}
```

*optimize images

* Test on different devices.

24) Document object Model (DOM) and Dynamic content:

The Document object Model (DOM) is a Programming interface that represents an HTML document as a tree of objects. Java script uses the DOM to access and modify webpage elements dynamically without reloading the page.

Role of DOM:-

* Access HTML elements.
* Modify text and styles.
* Add or remove elements.
* Respond to user actions.
* update content dynamically.

DOM Manipulation Example:- HTML

```
<P id:"message">Welcome</p>
<button onclick="changeText()">click Here</button>
```

Java script:-

```
function change Text(){
  document.getElementById("message").innerHTML="Welcome
  to Javascript!";
}
```

Another Example (Adding a New Item) HTML

```
<ul id="list">
<li>Apple</li>
</ul>
<button onclick="addItem()">Add</button>
```

Javascript:

```
function addItem(){
let li= document.createElement("li");
li.innerHTML="orange";
document.getElementById("list").appendchild(li);
}
```

Improvements for performance and user Experience:

* update only required DOM elements instead of reloding the page.
* Minimize unnecessary DOM manipulations.
* event delegation for better performance.
* Display loading indicators during data fetching.
* validate user input before updating the DOM.
* write clean and modular Javascript code.